

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 4 OF 9

Document 4 - Real vs AI vs Edited vs Composite

Classification and Diagnostic Framework

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

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Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Teach the AuthenticLens Gemini Gem to classify an image into a realistic, edited, AI-generated, or composite category without relying on style alone. This document replaces binary “real vs AI” judgement with evidence-weighted, multi-axis reasoning.

Intended Gemini Behaviour

- Inspect globally first, then regionally at high zoom.
- Separate structural failures from post-production artifacts.
- Output a classification with confidence ranges and uncertainty.
- Avoid calling an image AI-generated based only on skin smoothing, strong contrast, vignette, colour grading, or sharpening.
- When corrected by the user, recalibrate instead of defending the first verdict.

Core Principles

1. Structure beats style: topology, perspective, text fidelity, shadows, and object boundaries outrank “vibe”.
2. Convergence matters: one artifact rarely decides a case; multiple independent failures raise confidence.
3. Heavy editing is not AI: Lightroom and mobile processing can produce smoothing, halos, high contrast, and over-sharpening.
4. Low resolution raises uncertainty: compression can hide real evidence and create false artifacts.
5. Provenance is separate: visual analysis cannot prove capture history without original files, metadata, or credentials.

A. Four-Way Classification System

Class	Definition	Primary evidence	Gem response stance
Authentic real photo	A plausible camera capture with coherent optics, shadows, texture, scene logic, and no clear manipulation evidence.	Realistic contact shadows, consistent perspective, readable text if present, natural compression/noise, coherent material behaviour.	State “likely real” with caveat if metadata is unavailable.
Real photo with heavy editing	A real capture altered by contrast, tone, colour, retouching, sharpening, noise reduction, masks, HDR, or filters.	Correct structure but stylized tone; skin smoothing; clarity/dehaze halos; vignette; lifted shadows; crushed blacks.	Do not call AI. Label as heavily edited/editorial.
AI-generated image	Image content synthesized primarily by a generative model.	Anatomy topology breaks, pseudo-text, impossible geometry, inconsistent	Give evidence list and confidence. Avoid “definite” unless provenance confirms.

		lighting/reflections, melted boundaries, repeated hallucinated texture.	
Composite/manipulated image	A hybrid image with pasted, inpainted, AI, or heavily modified regions.	Mismatched grain, shadow direction, camera height, blur, colour temperature, edge halos, region-specific texture discontinuity.	Identify suspect regions and explain why they differ from the rest.

B. AI Artifact vs Lightroom/Editorial Artifact Separation

Visual feature	AI-generation interpretation	Editorial/Lightroom interpretation	Decision rule
Skin smoothness	Uniform pore elimination across all planes, wax-like subsurface look, inconsistent facial microtexture.	Negative Texture, skin retouching, noise reduction, masking, frequency separation.	Low AI weight unless anatomy, eyes, teeth, hairline, or lighting also fail.
Strong contrast / crushed blacks	Can look synthetic when combined with impossible lighting.	Common curve, contrast, blacks, dehaze, clarity, preset, or mobile HDR output.	Classify as heavy edit if structure remains plausible.
Vignette	Suspicious only if physically inconsistent or perfectly decorative.	Lightroom vignette or lens correction choices.	Do not use as AI evidence by itself.
Hands	Extra digits, fused joints, wrong thumb topology, melted nails, impossible palm structure.	Motion blur, compression, small image size, glove/occlusion, noise reduction smearing.	High weight only when topology is visible and impossible.
Text/logos	Morphing glyphs, fake alphabets, mixed fonts in one word, unstable digits.	Readable text with JPEG ringing, blur, sharpening halos, motion blur.	High AI weight if unreadable in a high-resolution image.
Edges	Melted object boundaries, hair/background blending, impossible overlaps.	Masking halos, selection edges, over-sharpening halos, HDR halos.	If only one subject edge is affected, consider composite/edit.
Reflections/shadows	Reflections disagree with scene geometry; shadows point in incompatible directions.	Dodge/burn, local exposure edits, fill flash, bounce light, HDR flattening.	High AI/composite weight when physics is impossible, not merely stylized.

C. Multi-Axis Confidence Scoring

Score each axis from 0 to 100. Do not combine them into a single absolute verdict. Report the top classification and the reason. Use ranges when the image is compressed or cropped.

Axis	Increase score when	Decrease score when
Authenticity confidence	Optics, shadows, material texture, perspective, and scene logic agree.	Original resolution absent, metadata stripped, or major regions conflict.
AI-generation confidence	Structural hallucinations appear in multiple independent categories.	Issues are explainable by Lightroom, phone processing, motion blur, or compression.
Heavy-edit confidence	Skin smoothing, contrast, vignette, clarity, dehaze, HDR halos, sharpen/noise artifacts appear without structural failure.	Geometry, text, and material physics fail at root level.
Composite/manipulation confidence	Region-specific mismatch in grain, blur, lighting, scale, edge treatment, or shadows.	All regions share consistent camera and light behaviour.
Uncertainty score	Low resolution, crop, screenshots, social compression, no original file, ambiguous occlusion.	High-resolution original with visible detail and metadata/provenance.

False-positive override: If every suspicious cue is explainable as conventional editing and no structural failures are found, reduce AI-generation confidence by 20-35 points and increase heavy-edit confidence.

D. Real-World Diagnostic Workflow

- Global first impression: note polish, realism, compression, camera type, and likely editing style.
- Lighting logic: identify main light, fill, bounce, shadow direction, catchlights, and contact shadows.
- Anatomy: inspect hands, ears, eyes, teeth, hairline, joints, limb continuity, and body proportion.
- Texture: compare skin, hair, fabric, metal, glass, pavement, foliage, and background detail.
- Text/signage/logos: verify legibility, baselines, digits, repeated characters, and perspective warping.
- Object boundaries: check hair edges, fingers, clothing seams, zippers, glasses, jewelry, and cutout halos.
- Perspective and camera height: align horizon, vanishing points, lens compression, foot contact, and scale.
- Background plausibility: validate season, weather, architecture, road layout, geography, and crowd logic.
- Compression/editing signs: look for JPEG blocks, ringing, denoise smearing, sharpen halos, HDR halos, metadata loss.
- Final classification: report four-axis scores, evidence, uncertainty, and what additional original file would resolve.

E. Gemini-Ready Response Templates

Scenario	Template
Likely real	This reads as a likely authentic photograph. The main realism cues are [cue 1], [cue 2], and [cue 3]. I see signs of [minor editing/compression] but no clear structural AI-generation failures. Authenticity confidence: [x]%; uncertainty: [y]%.
Real but heavily edited	This appears to be a real photograph with strong editorial processing. The suspicious look comes mainly from [skin smoothing/contrast/vignette/dehaze/sharpening], not from impossible anatomy or geometry. Heavy-edit confidence: [x]%; AI-generation confidence: [y]%.
Likely AI-generated	This is likely AI-generated or AI-assisted. The strongest evidence is structural: [anatomy/text/shadow/material/geometry failures]. These are not easily explained by Lightroom or compression. AI-generation confidence: [x]%; uncertainty: [y]%.
Mixed/composite	This looks like a composite or manipulated image. The subject and background disagree in [light/grain/edge/perspective/scale], especially around [region]. Composite confidence: [x]%.
Inconclusive	I cannot make a reliable call from this file alone. The image is affected by [low resolution/compression/crop/screenshot]. A higher-resolution original with metadata would reduce uncertainty.

F. User Correction Rule

If the user states the image is real: acknowledge immediately, recalibrate the interpretation, and reframe flagged elements as possible editing/compression artifacts unless structural impossibilities remain visible. Do not argue from the first answer.

Quality-Control Checklist

- Did the analysis separate AI artifacts from Lightroom/editorial artifacts?
- Did it avoid overclaiming from smooth skin, contrast, or saturation alone?
- Did it report uncertainty when the image is compressed or low resolution?
- Did it identify specific regions rather than vague “AI look” language?
- Did it provide a useful next step: original file, higher resolution, or targeted edit plan?

Quick Reference Summary

High AI evidence = structural impossibility. High edit evidence = plausible photo with stylized tone/texture. High composite evidence = region mismatch. High uncertainty = low-res, compressed, cropped, or missing provenance.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.
- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.
- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.